Air Quality Analysis: Model Comparison

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Introduction

- This project explores the relationship between meteorological variables and ozone levels using the 'airquality' dataset.

- Goal: Build and evaluate predictive models for ozone concentration based on weather-related variables.

Dataset Overview

- Dataset: R's built-in 'airquality' dataset
- Time: May–September 1973 | Location: New York City

- Variables:
- - Ozone (target)
- - Solar.R (radiation)
- - Wind (mph)
- - Temp (°F)
- - Month, Day

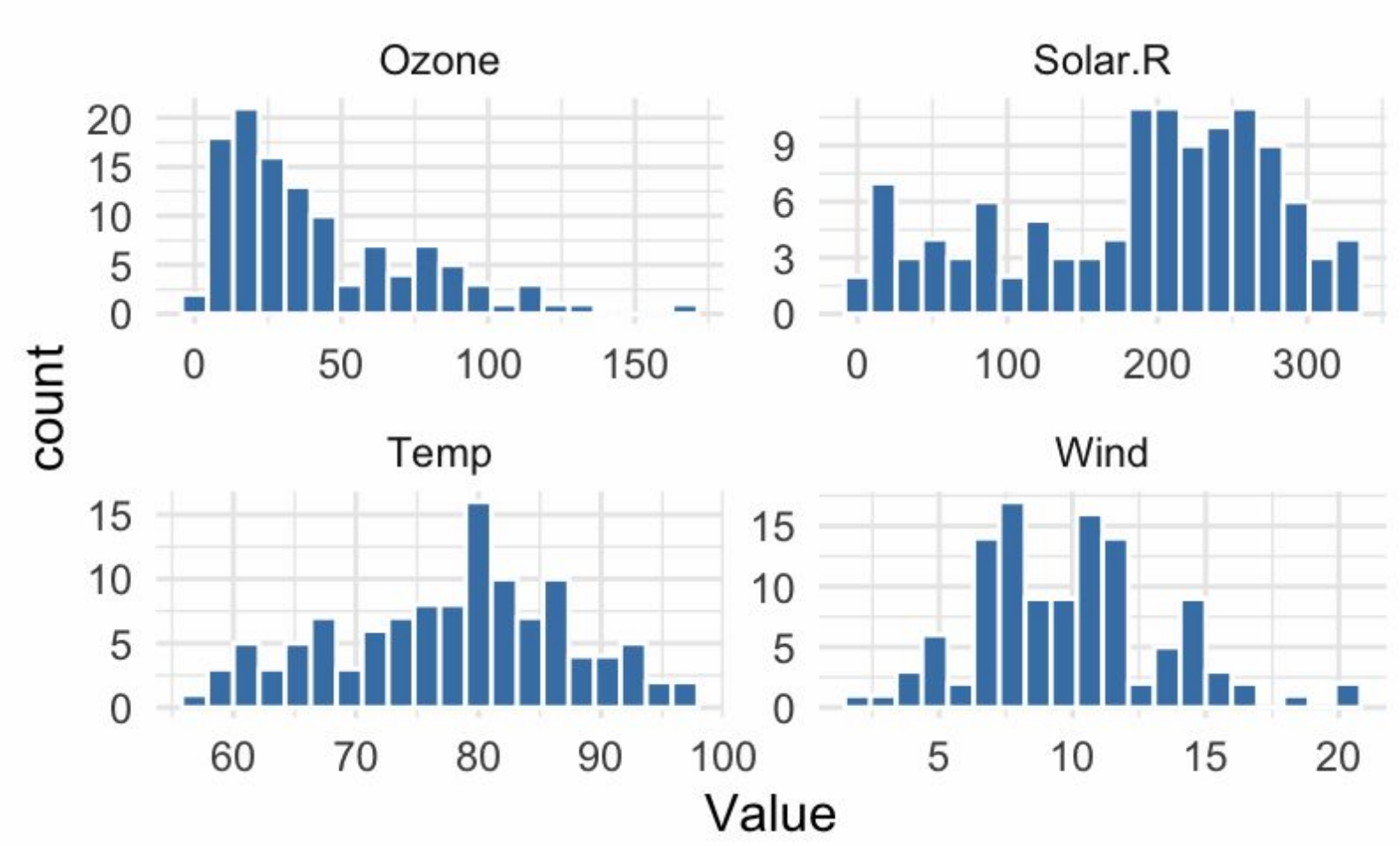
Data Cleaning

- - Removed 37 rows with missing Ozone
- - Imputed 7 missing Solar.R values with median
- - Standardized predictors for regularization models

Exploratory Analysis

- - Ozone is right-skewed; seasonal patterns observed
- - Correlation with Temp (~0.70), Wind (~-0.60), Solar.R (~0.35)
- - Highest Ozone: Summer months (June–August)

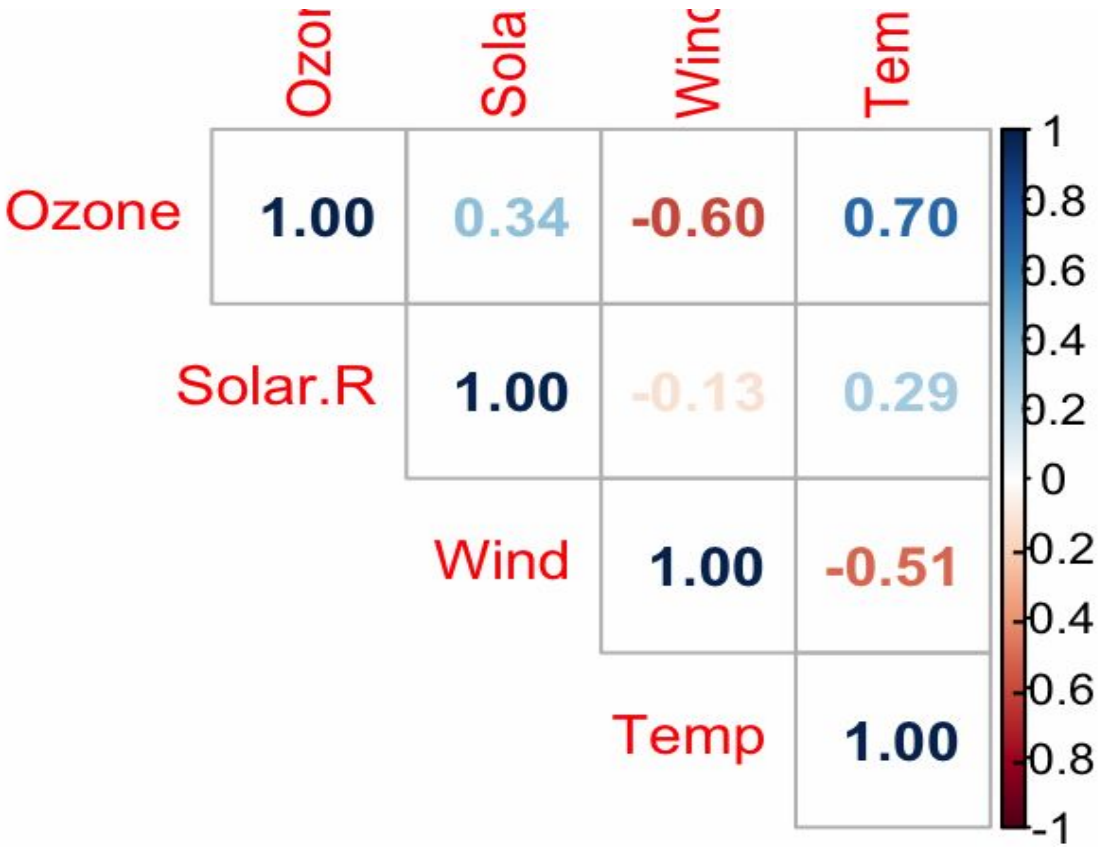
Histogram


Ozone
Solar.R
Temp
Wind
count
Value
20
15
10
5
0
0
50
100
150
9
6
3
0
0
100
200
300
15
10
5
0
60
70
80
90
100
15
10
5
0
5
10
15
20

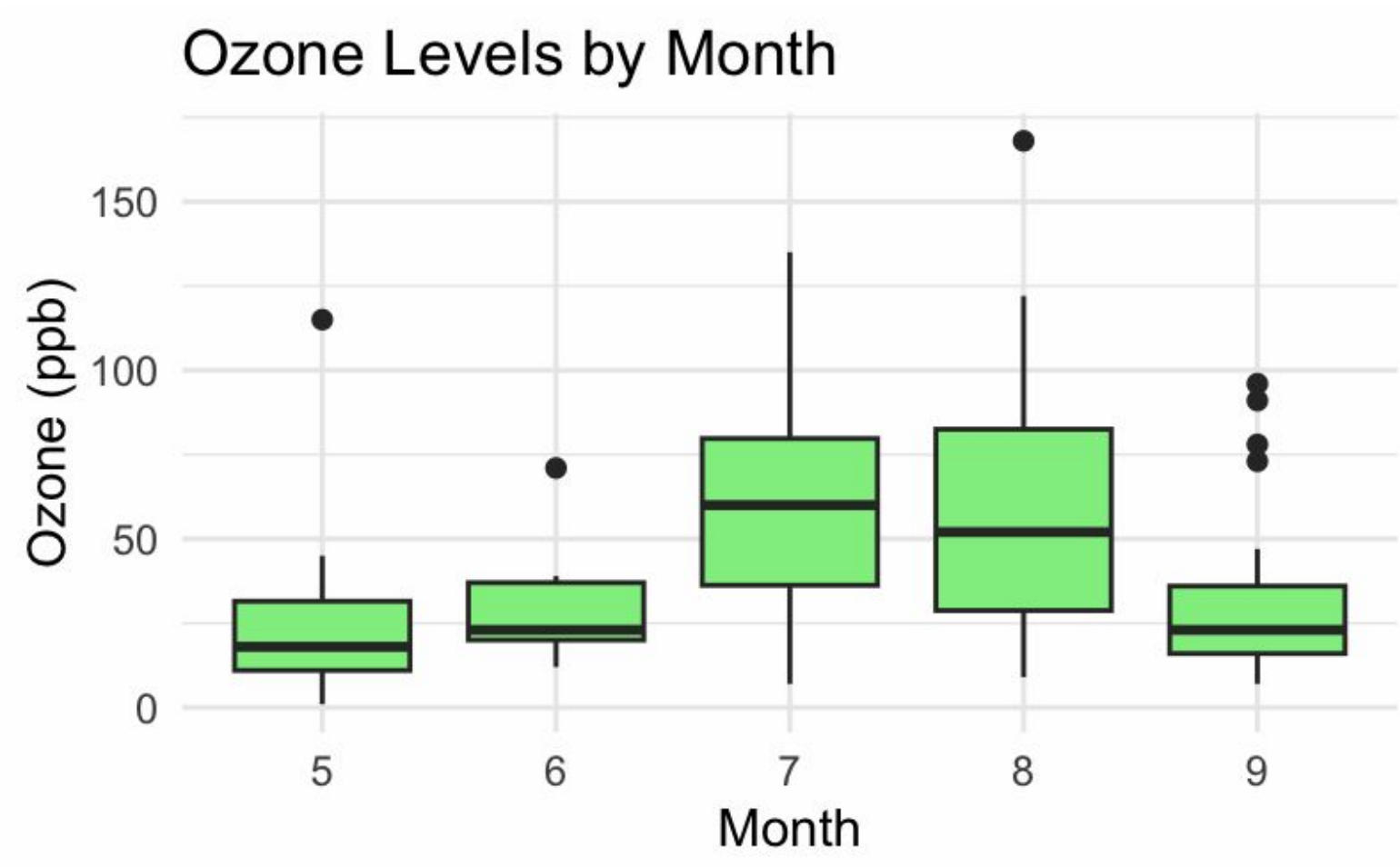
Models Selected

- Five regression models used:
- 1. Linear Regression
- 2. Ridge Regression
- 3. Lasso Regression
- 4. Decision Tree Regression
- 5. Random Forest Regression (Optional or placeholder)

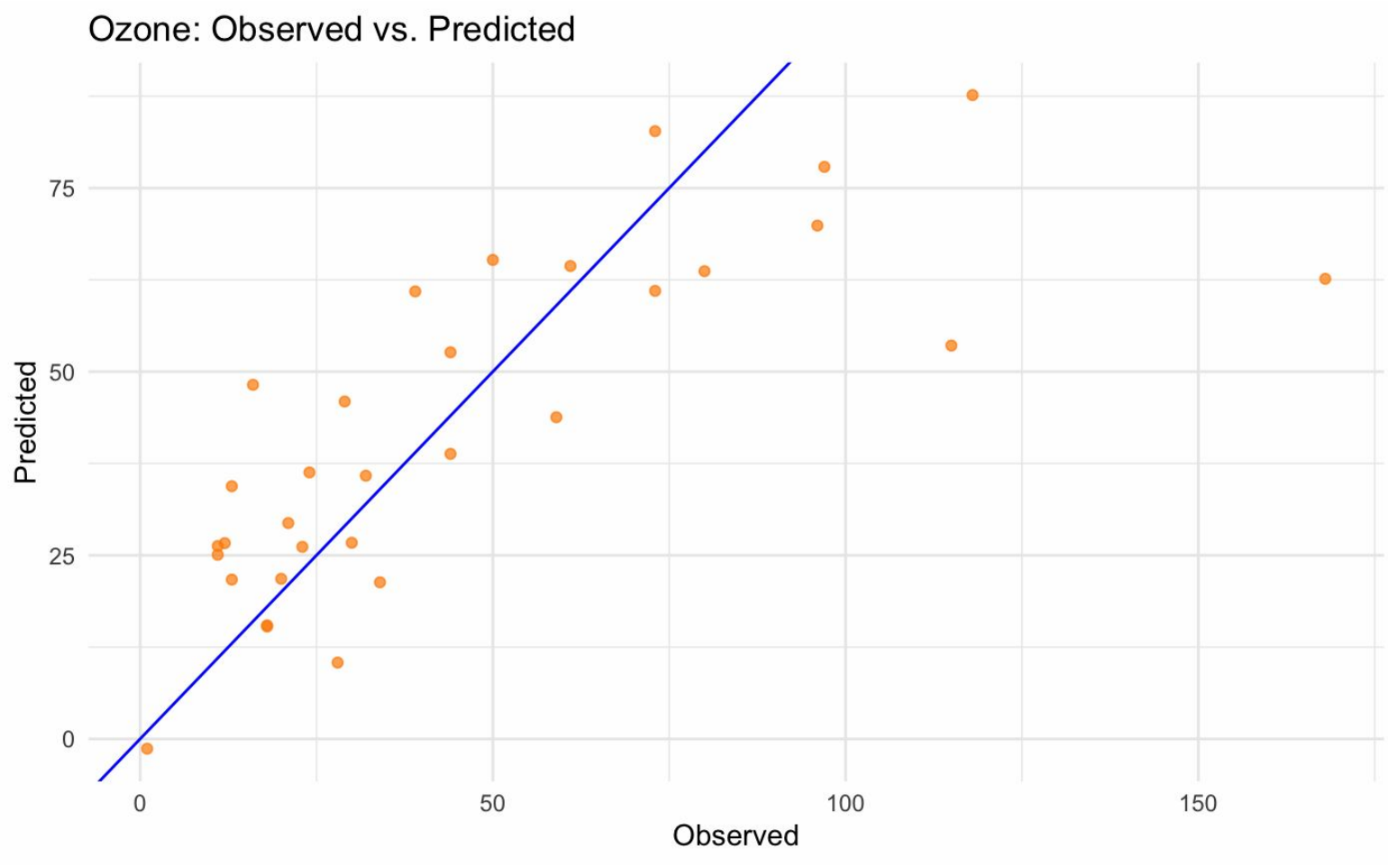
Correlation matrix:
Ozone correlates positively with Temp and negatively with Wind.



Ozone levels are highest in July and August, showing seasonal trends.



Observed vs. Predicted Ozone using Linear Regression. Moderate accuracy ($R^2 \approx 0.60$).

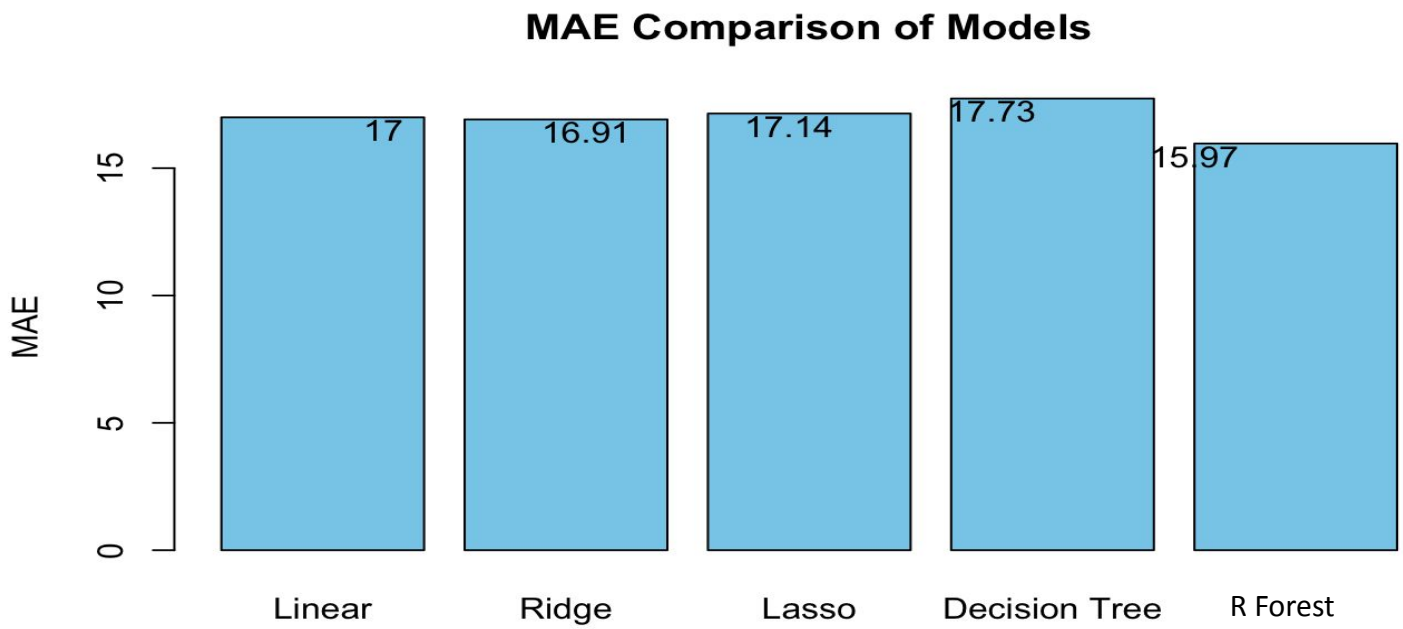


MAE Comparison of Models

- MAE for each model:
- - Linear: 17
- - Ridge: 16.91
- - Lasso: 17.14
- - Decision Tree: 17.73
- - Random Forest: 15.97

- $\longrightarrow$ Random Forest performed best overall.

MAE Comparison of Models



MAE comparison: Random Forest is the best performing model.

Model Interpretations

- - Linear: Coefficients indicate effect size
- - Ridge & Lasso: Regularize coefficients, reduce overfitting
- - Decision Tree: Easy to visualize, may overfit
- - Random Forest: Balances bias and variance well

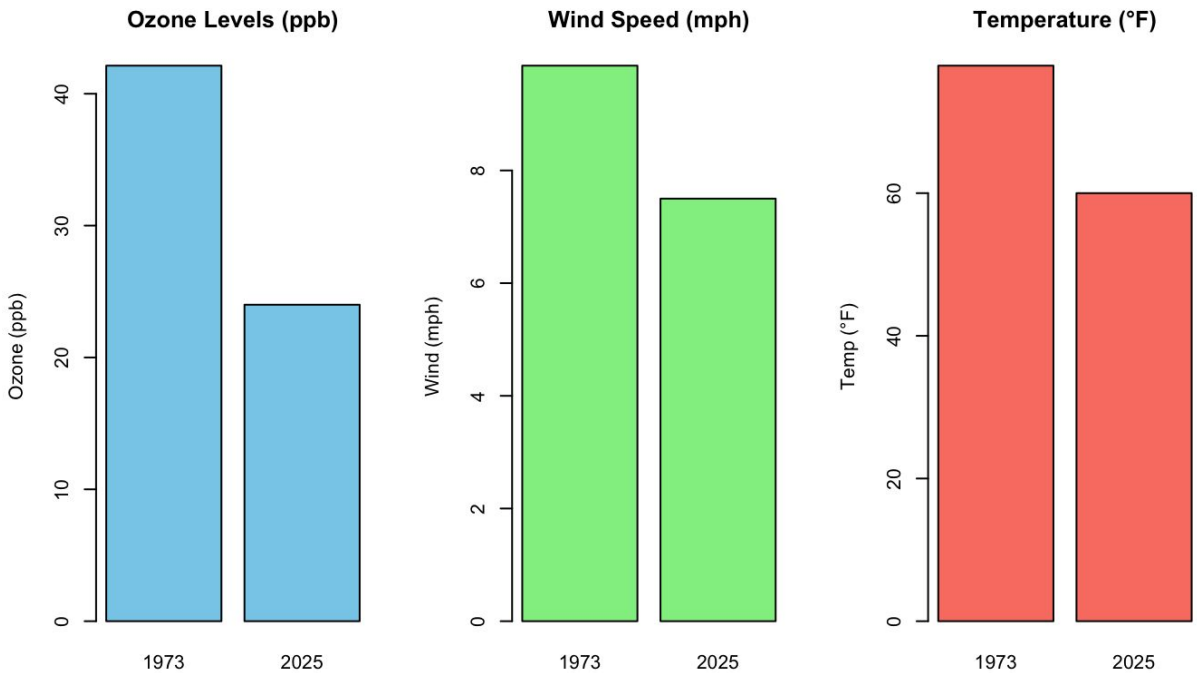
Conclusion (“airquality” 1973)

- **Random Forest** remains the most accurate model, with the **lowest MAE (15.97)**, confirming its strength in capturing nonlinear relationships and interactions between variables.
- **Temperature** continues to be the strongest predictor, showing consistent influence on ozone levels across all models.
- **Ridge Regression** (MAE = 16.91) performed slightly better than **Linear** and **Lasso**, showing that **regularization helps reduce overfitting** due to multicollinearity.
- **Decision Tree Regression** had the **highest MAE (17.73)**, suggesting that shallow trees may underperform without tuning.

Variable Comparison: 1973 vs 2025 (NYC)

Variable	1973	2025	Source
Ozone (ppb)	42.13	24 ppb	https://www.aqi.in/us/dashboard/united-states/new-york/new-york-city
Solar.R	185.9 (Langley)	N/A	N/A
Wind (mph)	9.86	7.5 mph	https://www.aqi.in/us/dashboard/united-states/new-york/new-york-city
Temp (°F)	77.87	60°F	https://www.aqi.in/us/dashboard/united-states/new-york/new-york-city

Air Quality Comparison: 1973 vs 2025 (New York City)



Comparison of average values between 1973 (airquality dataset) and recent 2025 data from NYC.

Ozone levels have decreased, while wind speed is slightly lower.

Current temperature is cooler, possibly due to seasonal variation.

Next Steps

- Perform **hyperparameter tuning** (especially for tree-based models).

- Test models on **other cities or updated air quality datasets** for generalization.

- Consider incorporating **additional features** (e.g., humidity, traffic data, time of day) to improve predictive performance.

Final Conclusion

- **Ozone levels in 2025 are significantly lower** than in 1973, indicating improvements in air quality policies and environmental awareness.

- **Random Forest provided the most accurate predictions** (lowest MAE), outperforming Linear, Ridge, and Lasso models.

- **Temperature remains the strongest predictor** of ozone levels, consistent across both historical and recent data.

- **Multicollinearity among predictors** (e.g., Temp, Wind, Solar.R) was effectively managed using **regularization techniques** like Ridge and Lasso.

- The observed vs. predicted ozone plot for linear regression showed **moderate accuracy ($R^2 \approx 0.60$)**, which suggests linear models capture part of the variation but not all.